

SymPy Bug Card

Bug 32: solveset misses a closed endpoint for $\exp(ix) = 1$

Task. Solve the periodic exponential equation over a closed real interval:

$$\text{solveset}(\exp(ix) = 1, x, [0, 2\pi]).$$

SymPy's answer.

$$\text{solveset}(\exp(ix) = 1, x, [0, 2\pi]) = \{0\}.$$

Correct answer.

$$\{0, 2\pi\}.$$

Explanation. The endpoint 2π is in the closed interval and $\exp(i2\pi) - 1 = 0$, so it exactly satisfies the equation. The returned set $\{0\}$ is therefore a strict subset of the intersection of the periodic solution family $2\pi\mathbb{Z}$ with $[0, 2\pi]$.

Likely root cause. The diagnosis traces the loss to `sympy/solvers/solveset.py`: for a real domain, `_invert` chooses `_invert_real`, which applies a single-valued logarithm to $\exp(ix)$ and discards the periodic family that complex inversion would produce. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-032-solveset-misses-a-closed-endpoint-for-exp-i-x-1/](#)